

APV-MaGa

Agrivoltaic Monitoring Platform

User Manual

Complete guide for administrators and agents

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1. Introduction to APV-MaGa

APV-MaGa (Agrivoltaic Monitoring and Management) is a real-time web monitoring platform for agrivoltaic installations in West Africa. It enables collecting, visualizing and exporting data from IoT sensors (ESP32) and manual field entries by agents.

Key Features:

- Real-time monitoring of solar, water, weather and agricultural data
- Dynamic REST API for ESP32/Arduino sensors
- Interactive dashboards with Chart.js graphs
- Manual field data entry
- CSV and Excel export by category or full site
- Multi-site and multi-user management with roles
- Responsive interface accessible on mobile and desktop

2. Login and Roles

2.1 Login

Access the platform via your browser. Enter your email address and password. Check 'Remember me' to stay logged in.

Login URL:

<https://your-domain.com/login>

2.2 User Roles

Role	Access	Permissions
Admin	Full	Manage sites, users, categories, parameters
Agent	Assigned sites	View, manual entry, export
Observer	Assigned sites	View and export only

3. Dashboard — Overview

The home page displays all sites you have access to with their key real-time indicators.

Interface elements:

Sidebar — List of your sites with colored status indicator (green=active, orange=alert)

Topbar — Title, UTC clock, period selector (1H/6H/24H/7D), CSV/Excel buttons

Site Cards — Key KPIs (solar power, borehole level, tank fill, temperature)

Sparkline — Mini-chart of last hour solar production

Cross-Site Trends — Comparative charts across all sites

Period Selector:

The 1H, 6H, 24H, 7D buttons at the top filter the data displayed on all charts.

4. Dashboard — Site Detail

Click on a site in the sidebar to access its detailed dashboard.

4.1 Category Tabs

Each site has tabs corresponding to its configured categories (Solar, Water, Irrigation, Weather, Agriculture, etc.).

4.2 KPI Cards

At the top of each tab, cards display the latest measured value for each parameter. Green cards indicate normal operation, orange cards indicate a warning.

4.3 Charts

Chart.js graphs display historical data for the selected period. Hover over the chart to see exact values.

4.4 Raw Data Tab

Displays all raw data grouped by timestamp. Filter by period (1H to 30D) and navigate by pages (20 records per page).

4.5 API Key Tab

Displays the site API key and available endpoints for connecting ESP32 sensors.

5. Manual Data Entry

The 'Manual Input' tab allows agents to enter field data that does not come from automatic sensors.

How to enter data:

1. Click the 'Manual Input' tab in the tab bar
2. Select the measurement date
3. Enter values for each parameter
4. Add optional notes
5. Click '+ Save' to save
6. The page automatically reloads on the Manual Input tab
7. KPI cards update with new values

Note: Manual parameters are identified by a blue 'Manual' badge on KPI cards and in Raw Data.

6. Raw Data & Export

6.1 Raw Data

The Raw Data tab displays all raw data for the site, grouped by timestamp and category.

- Period filter: 1H, 6H, 24H, 7D, 30D
- Grouped data — 1 row per sensor send
- Badge  Sensor (green) or  Manual (blue)
- Pagination: 20 records per page
- CSV/XLS export per category

6.2 Export

Button	Action
 CSV (category)	Exports category data as CSV
 XLS (category)	Exports category data as Excel
 All CSV	Exports all categories in one CSV file
 All Excel	Exports all categories as separate Excel sheets

7. ESP32 API — Sending Data

7.1 Available Endpoints

7.2 Authentication

All API requests require the site API key in the HTTP header:

Method	Endpoint	Description
POST	/api/sensors/{site}/{category}	Send sensor data
GET	/api/sensors/{site}/{category}/latest	Latest reading
GET	/api/sensors/{site}/status	Status of all categories

X-API-Key: apv-xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx

7.3 Arduino/ESP32 Example

```
#include <WiFi.h> #include <HTTPClient.h> void sendSolarData(float power, float irradiance) { HTTPClient http; http.begin("https://domain.com/api/sensors/site-slug/solar"); http.addHeader("X-API-Key", "apv-xxxxxxxx"); http.addHeader("Content-Type", "application/json"); String body = "{\"solar_output\":\"" + String(power) + "\", \"solar_irradiance\":\"" + String(irradiance) + "\"}"; int code = http.POST(body); http.end(); }
```

8. Administration

8.1 User Management

Admin Panel → Users. Create, approve, suspend or delete users.

- + **Add User** — Create a new user with role and status
- ✓ **Approve** — Approve a pending user
- **Suspend** — Temporarily suspend access
- **Activate** — Reactivate a suspended user
- ✕ **Delete** — Permanently delete (except admin)

8.2 Site Management

Admin Panel → Sites. Create and manage monitoring sites.

When creating a site, check 'Auto-create all categories' to automatically generate Solar, Water, Irrigation, Weather and Agriculture categories with their default parameters.

8.3 Categories & Parameters

Admin Panel → Sites → Categories → Parameters. Configure the categories and parameters for each site.

Field	Description
Name / Slug	Unique identifier (e.g. solar_output)
Input type	sensor (ESP32) or manual (agent)
Data type	float, integer, string, boolean
Unit	Measurement unit (kW, %, °C, m...)
Show on dashboard	Display as KPI card
Warning threshold	Value triggering Warning badge

9. FAQ

Q: My sensor sends data but it doesn't appear?

A: Check that the category and parameter slugs match exactly those configured in the Admin Panel. The API key must be for the correct site.

Q: How to add a new category?

A: Admin Panel → Sites → click 'Categories' for the site → '+ Add Category'. Then add parameters via 'Parameters'.

Q: Charts show 'No data'?

A: No data has been received for the selected period. Change the period (1H → 24H → 7D) or send data via the API.

Q: How to change a user's password?

A: The admin can create a new user. The user can change their password via My Profile → Change Password.

Q: The Manual Input button doesn't appear?

A: The Manual Input tab only appears if the site has at least one parameter with `input_type = 'manual'`. Configure it in Admin Panel → Parameters.

Q: How to connect multiple ESP32 to the same site?

A: Each ESP32 can send data to a different category using the same site API key. E.g: ESP32-1 → /solar, ESP32-2 → /water.